Zara Fashion E-commerce Management System

SQL Query Implementation

1.Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from the table.

Display all products:

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                          PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       101
Floral Dress
Zara                                               M
Black                             2499           1          1


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                          PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       102
Casual T-Shirt
Zara                                               L
White                              999           2          2
```

Display selected products:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 2000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                          PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       101
Floral Dress
Zara                                               M
Black                             2499           1          1


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                          PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       104
Running Shoes
Zara                                               8
White                             2999           4          4
```

1.2 WHERE Clause

The WHERE clause used to filter records based on a specified condition.

```
SQL> SELECT Product_ID, Product_Name, Brand, Price
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
BRAND                                              PRICE
-------------------------------------------------- ----------
       102
Casual T-Shirt
Zara                                                  999

       107
Oversized T-Shirt
Zara                                                 1199

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
BRAND                                              PRICE
-------------------------------------------------- ----------

       105
Leather Belt
Zara                                                 1299
```

1.3 ORDER Clause

The ORDER Clause used to sort the retrieved records in ascending or descending order.

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------
       109
Casual Sneakers
      3499

       104
Running Shoes
      2999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------

       106
Summer Dress
      2799

       101
Floral Dress
```

1.4 DISTINCT

The DISTINCT used to retrieve unique values without duplicates.

```
SQL> SELECT DISTINCT Category_ID
  2  FROM Product;

CATEGORY_ID
-----------
          1
          2
          3
          4
          5
```

2.Search products

2.1 Search products based on their price

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 1000 AND 3000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------
       101
Floral Dress
      2499

       103
Slim Fit Jeans
      1999

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
----------

       104
Running Shoes
      2999

       105
Leather Belt

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------
     PRICE
```

2.2 Search products based on category

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                         PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       101
Floral Dress
Zara                                               M
Black                                2499           1          1


PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                              PRODUCT_SIZE
-------------------------------------------------- ----------------
COLOUR                         PRICE CATEGORY_ID  SELLER_ID
------------------------------ ---------- ----------- ----------
       106
Summer Dress
Zara                                               S
Pink                                 2799           1
```

3.Retrieve customer information

```
SQL> SELECT * FROM Customer;

CUSTOMERID        AGE GENDER     PHONE
---------- ---------- ---------- ----------
EMAIL
--------------------------------------------------
ADDRESS
--------------------------------------------------
JOB
--------------------
       224          18 female     9840472788
aarthy.vsdvn@gmail.com
Banglore



CUSTOMERID        AGE GENDER     PHONE
---------- ---------- ---------- ----------
EMAIL
--------------------------------------------------
ADDRESS
--------------------------------------------------
JOB
--------------------
       108          19 female     9884752129
reshuu.23@gmail.com
Chennai
```

4.Applying filtering conditions

Displaying products with a price below 1000 or above 3000 using the OR operator.

```
SQL> SELECT Product_ID, Product_Name, Price, Category_ID
  2  FROM Product
  3  WHERE Price < 1000
  4  OR Price > 3000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       102
Casual T-Shirt
       999           2

       109
Casual Sneakers
      3499           4

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE CATEGORY_ID
```

Displaying products that does not belong to category ID 1

```
SQL> SELECT Product_ID, Product_Name, Price, Category_ID
  2  FROM Product
  3  WHERE Category_ID <> 1;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------
       102
Casual T-Shirt
       999           2

       103
Slim Fit Jeans
      1999           3

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE CATEGORY_ID
---------- -----------

       104
Running Shoes
      2999           4

       105
Leather Belt
```

5.Basic Business Reports

Business reports help analyse product prices,customer orders and product sales

5.1 Product Price Report

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE
----------
       109
Casual Sneakers
      3499

       104
Running Shoes
      2999

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE
----------

       106
Summer Dress
      2799

       101
Floral Dress

PRODUCT_ID
----------
PRODUCT_NAME
----------------------------------------
     PRICE
----------
      2499
```

5. 2. Customer Order Report

```
SQL> SELECT Order_ID, Customer_ID, Order_Date, Total_Amount
  2  FROM Orders;

  ORDER_ID CUSTOMER_ID ORDER_DAT TOTAL_AMOUNT
---------- ----------- --------- ------------
       101           1 01-OCT-26         4999
       102           2 02-OCT-26         2999
       103           1 03-OCT-26         1999
       104           3 04-OCT-26         5999
       105           2 05-OCT-26         3499
```

5. 3. Product Order Report

```
SQL> SELECT
  2      O.ORDER_ID,
  3      OD.PRODUCT_ID,
  4      P.PRODUCT_NAME,
  5      OD.QUANTITY,
  6      OD.UNIT_PRICE,
  7      OD.SUBTOTAL
  8  FROM ORDERS O
  9  JOIN ORDER_DETAILS OD
 10      ON O.ORDER_ID = OD.ORDER_ID
 11  JOIN PRODUCT P
 12      ON OD.PRODUCT_ID = P.PRODUCT_ID
 13  ORDER BY OD.SUBTOTAL DESC;

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
----------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------
       101        101
Floral Dress
         2       2499       4998

       104        109
Casual Sneakers
         1       3499       3499

  ORDER_ID PRODUCT_ID
---------- ----------
PRODUCT_NAME
----------------------------------------
  QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ----------

       105        104
Running Shoes
```

5.4 Customer-Wise Total Orders

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      COUNT(ORDER_ID) AS TOTAL_ORDERS
  4  FROM ORDERS
  5  GROUP BY CUSTOMER_ID
  6  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID TOTAL_ORDERS
----------- ------------
          1            2
          2            2
          3            1
```

5.5 Customer-Wise Total Purchase Amount

```
SQL> SELECT
  2      CUSTOMER_ID,
  3      SUM(TOTAL_AMOUNT) AS TOTAL_PURCHASE_AMOUNT
  4  FROM ORDERS
  5  GROUP BY CUSTOMER_ID
  6  ORDER BY TOTAL_PURCHASE_AMOUNT DESC;

CUSTOMER_ID TOTAL_PURCHASE_AMOUNT
----------- ---------------------
          1                  6998
          2                  6498
          3                  5999
```